MILITARY SPECIFICATION

RELAYS, ELECTROMAGNETIC, HYBRID, EMI SUPPRESSED, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 This specification covers EMI suppressed, hybrid, electromagnetic relays having integral solid-state drivers controlling the coil. These relays are designed for switching 230/400 volt 400 Hertz loads.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	- Solder; Tin Alloy: Lead-Tin Alloy; and Lead Alloy.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

MILITARY

MIL-I-10	- Insulating Compound, Electrical, Ceramic, Class L.
MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116	- Preservation, Methods of.
MIL-W-583	- Wire, Magnet, Electrical.
MIL-P-997	- Plastic Material, Laminated, Thermosetting, Electrical Insulation: Sheets, Glass Cloth, Silicone Resin.
MIL-S-5002	- Surfaces Treatments and Inorganic Coatings for Metal Surfaces of Weapon Systems.
MIL-W-5088	- Wiring, Aircraft, Installation of.
MIL-W-6858	- Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat Resisting Alloys, and Titanium Alloys, Spot and Seam.
MIL-S-6872	- Soldering Process, General Specification For.
MIL-W-6873	- Welding, Flash, Carbon and Alloys Steel.
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys.
MIL-W-8604	- Welding of Aluminum Alloys, Process for.
MIL-W-8611	- Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for.
MIL-P-15037	- Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-P-15047	- Plastic Sheets, Laminated Thermosetting, Nylon Fabric Base, Phenolic-Resin.
MIL-W-18326	- Welding of Magnesium Alloys, Gas and Arc, Manual and Machine, Processes for.
MIL-G-45204	- Gold Plating - Electrodeposited.
MIL-C-45662	- Calibration System Requirements.

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MIL-R-83372/1	-	Relay, Electromagnetic, Hybrid, EMI Suppressed, 4PDT, 5 Amperes, 230/400 V, 400 Hz.
MIL-R-83372/2	-	Relay, Electromagnetic, Hybrid, EMI Suppressed, 2PDT, 5 Amperes, 230 V, 400 Hz.
MIL-R-83372/3	-	Relay, Electromagnetic, Hybrid, EMI Suppressed, 3PST, 10 Amperes, 230/400 V, 400 Hz.
MIL-R-83372/4	-	Relay, Electromagnetic, Hybrid, EMI Suppressed, 3PST, 10 Amperes, 230/400 V, 400 Hz, With Auxiliary 1PDT Contacts.

STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads on 40" x 48" Pallets.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MS20659	-	Terminal, Lug, Crimp Style, Copper, Uninsulated, Ring Tongue, Type I, Class 1.
MS25036	-	Terminal, Lug, Crimp Style, Copper Insulated, Ring Tongue, Bell-Mounted, Type II, Class 1.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

American Society for Testing and Materials (ASTM)

D1868-61T - Method for Corona Measurement.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Society of Automotive Engineers, Inc.

AMS 3332 - Silicone Rubber - Extreme Low Temperature Resistant 15 - 30.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, New York 10017.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern (see 6.2).

3.2 First article. First article relays shall be furnished for the inspections specified in 4.5.

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3.3 Materials. Materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the relays to meet the performance requirements of this specification. Materials used shall be self-extinguishing and shall not support combustion; give off gases in quantities sufficient to cause explosion of sealed enclosures; cause contamination to any part of the relay; or form current-carrying tracks when subjected to any of the tests specified herein. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product. (See 6.3.)

3.3.1 Plastic. Laminated plastic material shall conform to MIL-P-997, MIL-P-15037, or MIL-P-15047. Molded plastic material shall conform to MIL-M-14. Cotton-filled or wood-flour-filled materials shall not be used.

3.3.2 Ceramic. Ceramic insulating material shall conform to MIL-I-10, grade L422 or higher. Ceramic used for external surfaces shall be glazed.

3.3.3 Metals. Metals shall be of a corrosion-resistant type, or shall be treated to resist corrosion.

3.3.3.1 Zinc and cadmium. Zinc and cadmium shall not be used.

3.3.3.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not touch each other. (Dissimilar metals are as defined in MIL-STD-889.)

3.3.3.3 Titanium. Contact of silver with titanium or its alloys is prohibited. Contact of halogenated solvents or methanol with titanium or its alloys is prohibited.

3.3.4 Magnet wire. Magnet wire shall conform to MIL-W-583.

3.3.5 Rubber. Rubber shall conform to Aerospace Material Specification (AMS) 3332.

3.3.6 Solder. Unless otherwise specified (see 3.1), compositions Sn70, Sn60, and Sb5 solder, conforming to QQ-S-571, shall be used.

3.3.6.1 Soldering processes. Soldering processes shall conform to MIL-S-6872. Soldering shall be so executed that both a positive electrical and strong mechanical connection is assured. Joints shall be mechanically secure before soldering and shall not depend upon solder alone for mechanical strength.

3.4 Design and construction. Relays shall be of the design, construction, weight, and physical dimensions specified (see 3.1). Unless otherwise specified (see 3.1), relays shall be designed for continuous duty at rated voltages and currents.

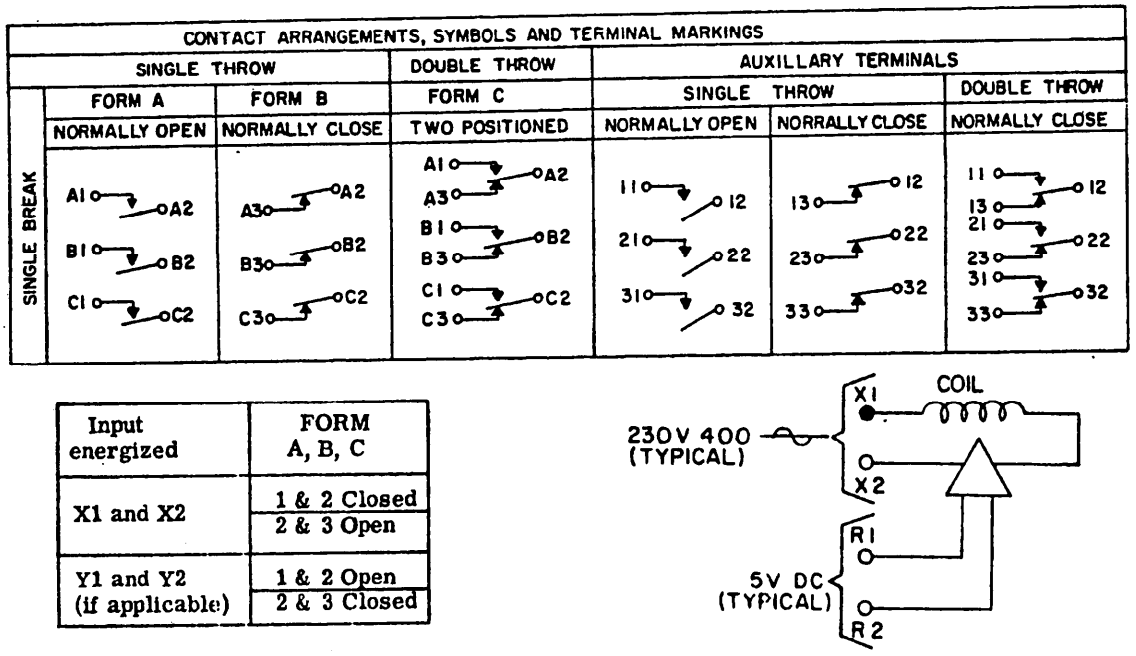
3.4.1 Mounting means. Relay mounting means (e.g., brackets, studs, and flanges) shall be electrically connected to the case and shall be electrically conductive over all areas which can mate to other mounting surfaces.

3.4.1.1 Mounting brackets. When applicable (see 3.1), mounting brackets shall be an integral part of the relay case.

3.4.2 Case. Relays shall be hermetically sealed in metal cases. The case shall be of sufficient mechanical strength to withstand the normal abuse incurred in handling, transit, storage, and installation without causing malfunction or distortion of parts. The case shall not be used for internal grounding, nor may it be used as part of the magnetic circuit. (See 6.4.)

3.4.3 Terminals. Terminals shall be arranged and marked as specified (see 3.1 and figure 1). Screw-type terminals shall be as specified (see 3.1). Plug-in type terminals shall mate with the specified socket contacts; plating shall be in accordance with MIL-G-45204, type II, class 2.

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Input energized	FORM A, B, C
X1 and X2	1 & 2 Closed 2 & 3 Open
Y1 and Y2 (if applicable)	1 & 2 Open 2 & 3 Closed

INPUT CIRCUIT

NOTES:
1. Contacts are shown with the input circuit(s) de-energized.
2. As a result of input energization, the open and closed positions are as above.
3. Contact identifications are typical.

FIGURE 1. Symbols and markings for contacts and inputs.

3.4.4 Input. The coil and driver ratings and characteristics shall be as specified (see 3.1).

3.4.5 Contacts. The contact rating(s) shall be as specified (see 3.1).

3.4.5.1 Contact arrangement. Contact arrangements shall be in accordance with figure 1, and shall be as specified (see 3.1).

3.4.5.2 Contact action. Relays shall be so designed that if a normally open contact is closed (as by contact welding), no normally closed contact can close; and conversely, if a normally closed contact is closed, no normally open contact can close.

3.4.5.3 Contact springs. Contact springs shall be of a non-corrosive material and may be plated, except zinc, cadmium, or tin plating shall not be used.

3.4.6 Threaded parts. All threaded parts shall be in accordance with Handbook H28, and shall be as specified (see 3.1).

3.4.7 Welding and brazing. Welding and brazing methods shall be in accordance with MIL-W-6858, MIL-W-6873, MIL-B-7883, MIL-W-8604, MIL-W-8611, or MIL-W-18326, as applicable.

3.4.8 Protective treatment. Materials subject to deterioration or corrosion during service shall be protected in accordance with MIL-S-5002.

3.5 Performance.

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3.5.1 Seal. When relays are tested as specified in 4.7.2, the maximum total leakage of gas atmosphere shall not exceed a rate of 1×10^{-6} atmospheric cubic centimeters per second (atm cc/sec) per cubic inch of net sealed volume (for relays over 2 cubic inches in volume) or 1×10^{-8} atm cc/sec (for relays of 2 cubic inches net sealed volume or less).

3.5.2 Terminal strength. When relays are tested as specified in 4.7.3, there shall be no evidence of loosening or breaking of the terminals; and there shall be no deformation to the threads of screw terminals; no bending (see NOTE) of a pin or damage to the insulating base of plug-in relays; nor shall there be any other damage which would adversely affect the normal operation of the relay.

NOTE: Bending of plug-in terminals shall not be construed as damage provided they can be re-formed in a manner to permit proper mating with the applicable socket.

3.5.3 Insulation resistance. When relays are tested as specified in 4.7.4, the insulation resistance shall be as specified (see 3.1).

3.5.4 Dielectric withstanding voltage. When tested as specified in 4.7.5, relays shall withstand the test voltage specified without damage, and there shall be no leakage current in excess of 1.0 milliampere (mA) nor evidence of damage due to arcing (air discharge), flashover (surface discharge), or insulation breakdown (puncture discharge). (NOTE: Corona alone shall not be considered a breakdown.)

3.5.5 Contact resistance or voltage drop. When relays are tested as specified in 4.7.6, the voltage drop across each contact shall not exceed the applicable value specified in table I.

TABLE I. Contact resistance or voltage drop.

Contact type	Before load/endurance test		During and after load/endurance test	
	Average reading	Maximum individual reading	Average reading	Maximum individual reading
Main - - - - - - - - - - -	0.150 volt	0.175 volt	0.175 volt	0.200 volt
Auxiliary - - - - - - - -	---	.05 ohm	---	0.1 ohm
Status indicator - - - - -	---	0.1 volt	---	0.1 volt

3.5.6 Status indication (when specified, see 3.1). When relays are tested as specified in 4.7.7, the voltage drop across a closed contact shall not exceed 0.1 volt and the current through an open contact circuit shall not exceed 0.1 mA.

3.5.7 Driver input and coil currents. When relays are tested as specified in 4.7.8, both the coil current and the driver current shall be as specified (see 3.1).

3.5.8 Operate voltage. When relays are tested as specified in 4.7.9, all switching circuits shall make positive contact or open, as applicable.

3.5.9 Release voltage. When tested as specified in 4.7.10, relays shall return to the de-energized position within the specified voltage range (see 3.1).

3.5.10 Contact bounce, operate, and release times. When relays are tested as specified in 4.7.11, the contact bounce, operate, and release times shall be within the limits specified (see 3.1). The operate and release times shall not include the contact bounce time. Contacts shall show no evidence of any normally open contact closing before all normally closed contacts open, nor any normally closed contact making before all normally open contacts open. Unless otherwise specified, open contacts shall close within 2 milliseconds (ms) of each other, and closed contacts shall open within 2 ms of each other.

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3.5.11 Transient voltage. When relays are tested as specified in 4.7.12, there shall be no opening of closed contacts in excess of 10 microseconds (μs) nor closing of open contacts in excess of 10 μs.

3.5.12 Electric power dissipation. When relays are tested as specified in 4.7.13, the total power shall not exceed the value specified (see 3.1).

3.5.13 Acoustical noise. When relays are tested as specified in 4.7.14, there shall be no opening of closed contacts in excess of 10 μs nor closing of open contacts in excess of 10 μs. There shall be no evidence of damage and relays shall meet the following post-test requirements:

Dielectric withstanding voltage	Shall be as specified in 3.5.4.
Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.

3.5.14 Thermal shock. When relays are tested as specified in 4.7.15, there shall be no breaking, cracking, chipping, or flaking of the finish, nor loosening of the terminals; and relays shall meet the following post-test requirements:

Insulation resistance	Shall be as specified in 3.5.3.
Dielectric withstanding voltage	Shall be as specified in 3.5.4.
Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.
Contact bounce, operate, and release times	Shall be as specified in 3.5.10.

3.5.15 Temperature-altitude. When relays are tested as specified in 4.7.16, there shall be no breaking, cracking, chipping, or flaking of the finish, nor loosening of the terminals; and relays shall meet the following requirements during the test and as post-test requirements:

Operate voltage	Shall be as specified (see 3.1).
Release voltage	Shall be as specified (see 3.1).
Contact bounce, operate, and release times	Shall be as specified in 3.5.10.
Contact resistance or voltage drop	Shall be as specified in 3.5.5.
Insulation resistance	Shall be as specified in 3.5.3 (post-test only).
Dielectric withstanding voltage	Shall be as specified in 3.5.4.

3.5.16 Continuous current. When tested as specified in 4.7.17, relays shall remain mechanically and electrically operative and shall show no evidence of damage such as loosening of terminals. Relays shall meet the following requirements:

Operate voltage	Shall be as specified (see 3.1).
Release voltage	Shall be as specified (see 3.1).
Contact resistance or voltage drop	Shall be as specified in 3.5.5.
Contact bounce, operate, and release times	Shall be as specified in 3.5.10.
Insulation resistance	Shall be as specified in 3.5.3.
Dielectric withstanding voltage	Shall be as specified in 3.5.4.
Driver input and coil currents	Shall be as specified in 3.5.7.

3.5.17 Shock (specified pulse). When relays are tested as specified in 4.7.18, there shall be no closing of open contacts in excess of 10 μs nor opening of closed contacts in excess of 100 μs. There shall be no evidence of structural failure, loss of seal, or other damage which might impair the operation of the relay. Bistable relays shall remain in their last energized position with no voltage applied to the input(s). Relays shall meet the following post-test requirements:

Dielectric withstanding voltage	Shall be as specified in 3.5.4.
Contact resistance or voltage drop	Shall be as specified in 3.5.5.
Contact bounce, operate, and release times	Shall be as specified in 3.5.10.

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3.5.18 Random vibration. When relays are tested as specified in 4.7.19, there shall be no opening of closed contacts in excess of 10 μs nor closing of open contacts in excess of 10 μs. Relays shall meet the following post-test requirements:

Dielectric withstanding voltage	Shall be as specified in 3.5.4.
Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.
Contact bounce, operate, and release times	Shall be as specified in 3.5.10.

3.5.19 Electromagnetic interference and compatibility (EMIC). When tested as specified in 4.7.20, relays shall not generate electromagnetic interference in excess of the limits specified in MIL-STD-461.

3.5.20 Case-to-ground resistance. When relays are tested as specified in 4.7.21, the resistance between the case and mounting member shall not exceed 0.1 ohm.

3.5.21 Compartment pressure. When tested as specified in 4.7.22, relays shall show no evidence of damage and shall meet the following post-test requirements:

Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.
Dielectric withstanding voltage	Shall be as specified in 3.5.4.

3.5.22 Explosive decompression. When tested as specified in 4.7.23, relays shall show no evidence of damage and shall meet the following post-test requirements:

Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.
Dielectric withstanding voltage	Shall be as specified in 3.5.4.

3.5.23 Acceleration. When relays are tested as specified in 4.7.24, the contacts shall remain in the relased position with no voltage applied to the input, and shall remain in the operated position when the inputs are energized. Bistable relays shall remain in the position attained with last energized. Relays shall meet the following requirements:

Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.
Dielectric withstanding voltage	Shall be as specified in 3.5.4.

Relays shall show no evidence of damage.

3.5.24 Corona. When relays are tested as specified in 4.7.25, there shall be no indication of corona, as exhibited by spikes or other irregularities appearing on the oscilloscope screen.

3.5.25 Rupture. When relays are tested as specified in 4.7.26, there shall be no evidence of electrical failure (such as contact welding, failure of the contacts to make or break the test current, or blowing of the fuse connecting case to system ground) and the contact resistance or voltage drop shall be as specified in 3.5.5.

3.5.26 Load/endurance. When relays are tested as specified in 4.7.27, there shall be no misses no evidence of loosening of the terminals, and the case to ground fuse shall remain electrically continuous. Relays shall meet the following requirements:

Contact resistance or voltage drop	Shall be as specified in 3.5.5.
Driver input and coil currents	Shall be as specified in 3.5.7.
Insulation resistance	Shall be as specified in 3.5.3.
Dielectric withstanding voltage	Shall be as specified in 3.5.4.
Operate voltage	Shall be as specified in 3.5.8.
Release voltage	Shall be as specified in 3.5.9.
Contact bounce, operate, and release times	Shall be as specified in 3.5.10.

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3.6 Marking. Relays shall be marked in accordance with method I of MIL-STD-1285, and shall include the military part number; source code or manufacturer's name; date code and lot symbol; circuit schematic diagram; highest contact rating (amperes) and rated input voltage; and dc or frequency as applicable.

3.7 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) First article inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabricating the relays, are in accordance with the applicable referenced documents prior to such fabrication.

TABLE II. Materials inspection.

Materials	Requirement paragraph	Applicable documents
Plastic- - - - - - - - -	3.3.1	MIL-P-997, MIL-P-15037, MIL-P-15047, MIL-M-14
Insulating materials- -	3.3.2	MIL-I-10
Magnet wire- - - - - -	3.3.4	MIL-W-583
Rubber - - - - - - - - -	3.3.5	AMS 3332
Solder - - - - - - - - -	3.3.6	QQ-S-571

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4.1 Load conditions during tests. The loads of the relay being tested shall be connected to the grounded side of their power supply; and each contact shall be connected to an individual load. All tests during which the contacts are loaded and being cycled shall be conducted with the case of the relay connected to the power supply ground or neutral through a normal blow fuse rated at 5 percent of the contact load, maximum, but not less than 0.100 ampere or more than 3 amperes. For relays with nongrounded case ratings, tests for isolated-case ratings may be made with the case electrically isolated from the power supply ground.

4.4.2 Monitoring devices. Monitoring devices shall not load the contacts above their lowest rating.

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4.4.3 Power supplies. Unless otherwise specified herein, power supplies shall have no more than 5 percent regulation over the range from 0 to twice applicable load current. A dc power supply shall have no more than 5 percent voltage ripple. An ac power supply shall be within 1 percent of the specified frequency and shall be sinusoidal with a form factor between 0.95 and 1.25.

4.4.4 Grounding. Unless otherwise specified herein, one side of a single-phase power supply shall be grounded, the neutral of a three-phase power supply shall be grounded, and one side of a dc power supply shall be grounded.

4.4.5 Loads. Test loads and circuits shall be so arranged that the specified current will flow through each pole. Line-to-line and line-to-ground voltages shall be as specified (see 3.1). The load test cycles shall be performed in any number of continuous periods, each not less than 3 hours. The relays shall be suspended or mounted in still air by a thermal insulating material. When testing multipole relays with three-phase ratings, three-phase loads shall be connected to adjacent contacts. During endurance tests, every operation of each contact shall be monitored for failure to make, carry, and break the specified load. The minimum sensing period shall be 10 percent of the dwell time in the open or closed position. Test equipment must either lock in the state of failure or record the sequence number of the miss. Relays which pass three-phase load tests need not undergo corresponding single-phase load tests if the single-phase current and voltage ratings do not exceed the three-phase current and line-to-neutral voltage ratings.

4.4.6 Wire. In any of the specified load tests (see 4.7.27), each individual wire shall be a minimum of 3 feet in length and of an applicable size conductor (copper) for single use in free air as listed in MIL-W-5088 determined by the rated resistive load of the relay. If the relay rating does not coincide with wire size, the next larger diameter wire shall be used.

4.4.6.1 Terminal lugs. Wire shall be terminated with an applicable size and type of terminal lug according to MS20659 or MS25036, or other currently approved military standard.

4.4.7 Terminal covers and barriers. When terminal covers or barriers, or both, are specified (see 3.1), all load and environmental tests shall be conducted with the terminal covers or barriers, or both, in place.

4.4.8 Post-tests. A test procedure (e.g., dielectric withstanding voltage) required at the conclusion of a procedure shall be known as a post-test of that procedure. Some post-tests are required to be repeated on sample units as post-tests of more than one environmental test procedure (4.7.14 thru 4.7.25). Such a post-test shall be required only at the conclusion of the last environmental test procedure requiring it.

4.5 First article inspection. First article inspection shall be performed by the supplier, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders. (See 6.5.)

4.5.1 Sample size. Five relays shall be selected at random, numbered serially 1 through 5, and subjected to the inspection as specified in table III.

4.5.2 Inspection routine. Sample relays shall be subjected to the inspections indicated with an "X" and in the order and sequence shown in table III.

4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval, except when one sample unit fails the life test, a duplicate sample unit may be subjected to the inspection routine and no failures shall be allowed.

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TABLE III. First article inspection.

Examination or test	Requirement paragraph	Test paragraph	Sample unit 1	2	3	4	5
Visual and mechanical examination	3.1, 3.3, 3.4, 3.6, and 3.7	4.7.1	X	X	X	X	X
Seal	3.5.1	4.7.2	X	X	X	X	X
Terminal strength	3.5.2	4.7.3			X	X	
Insulation resistance	3.5.3	4.7.4	X	X	X	X	X
Dielectric withstanding voltage	3.5.4	4.7.5	X	X	X	X	X
Operating characteristics:							
Contact resistance or voltage drop	3.5.5	4.7.6	X	X	X	X	X
Status indication (when specified, see 3.1)	3.5.6	4.7.7	X	X	X	X	X
Driver input and coil currents	3.5.7	4.7.8	X	X	X	X	X
Operate voltage	3.5.8	4.7.9	X	X	X	X	X
Release voltage	3.5.9	4.7.10	X	X	X	X	X
Contact bounce, operate, and release times	3.5.10	4.7.11	X	X	X	X	X
Transient voltage	3.5.11	4.7.12	X	X	X	X	X
Electric power dissipation	3.5.12	4.7.13	X	X			
Acoustical noise	3.5.13	4.7.14	X	X			
Thermal shock	3.5.14	4.7.15	X		X		
Temperature-altitude	3.5.15	4.7.16		X	X		
Continuous current	3.5.16	4.7.17					X
Shock (specified pulse)	3.5.17	4.7.18	X			X	
Random vibration	3.5.18	4.7.19		X		X	X
Electromagnetic interference and compatibility	3.5.19	4.7.20		X			
Case-to-ground resistance	3.5.20	4.7.21					X
Compartment pressure	3.5.21	4.7.22	X				X
Explosive decompression	3.5.22	4.7.23				X	X
Acceleration	3.5.23	4.7.24			X	X	
Corona	3.5.24	4.7.25			X	X	
Load/endurance	3.5.26	4.7.27					
Rupture	3.5.25	4.7.26	X	X	X		X
Inductive load life 1/	3.5.26	4.7.27.1	X			X	
Motor load life 1/	3.5.26	4.7.27.2			X		
Resistive load life 1/	3.5.26	4.7.27.3		X			
Lamp load life 1/	3.5.26	4.7.27.4					X
Seal	3.5.1	4.7.2	X	X	X	X	X
Visual and mechanical examination	3.1, 3.3, 3.4, 3.6, and 3.7	4.7.1	X	X	X	X	X

1/ If a relay is not rated for all types of the listed specified test loads, it shall be tested for a load for which it is rated using this order of precedence: Inductive, motor, resistive.

4.6 Quality conformance inspection. Inspection of product for delivery shall be as specified in table IV, in the order shown.

TABLE IV. Quality conformance inspection.

Examination or test	Requirement paragraph	Test paragraph
Visual and mechanical examination	3.1, 3.3, 3.4, 3.6, and 3.7	4.7.1
Seal	3.5.1	4.7.2
Insulation resistance	3.5.3	4.7.4
Dielectric withstanding voltage	3.5.4	4.7.5
Contact resistance or voltage drop	3.5.5	4.7.6
Driver input and coil currents	3.5.7	4.7.8
Operate voltage	3.5.8	4.7.9
Release voltage	3.5.9	4.7.10
Contact bounce, operate, and release times	3.5.10	4.7.11

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at each specified voltage require testing of each circuit

4.6.1 Inspection of preparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Methods of examination and test. Testing of dual coil relays shall be repeated with each coil serving in turn as the operating coil. Testing of bistable relays shall be repeated with each position serving in turn as the operated position. Unless otherwise specified herein, relays shall be operated by applying rated operate voltages (see 3.1) to the driver input and coil circuits.

4.7.1 Visual and mechanical examination. Relays shall be examined to verify that the materials, design, construction, physical dimensions, weight, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.6, and 3.7). Visual and mechanical examination shall be performed with professional quality measuring equipments to determine dimensions, weights, and surface defects.

4.7.2 Seal (see 3.5.1). Relays shall be tested in accordance with method 112 of MIL-STD-202. The following details shall apply:

(a) Test-condition letter - C (procedure III or IV at the option of the manufacturer). Silicone oil shall not be used.
(b) Leakage-rate sensitivity - 1×10^{-8} atm cc/sec.
(c) Reduced pressure of chamber and duration of pressurization (procedure IV) - Below 5 millimeters of mercury (mmHg) until leak detector reading is stabilized.
(d) Measurements after test - Not applicable.

4.7.3 Terminal strength (see 3.5.2). Relays shall be tested in accordance with method 211 of MIL-STD-202, in accordance with the following, as applicable.

4.7.3.1 Pull test (all terminal types). Terminals shall be tested as specified in test-condition A. For screw terminals, the force shall be as specified in table V. For other types of terminals, the force shall be 10 pounds.

TABLE V. Terminal strength test details.

Screw size	Force	Torque
	pounds	pound-inches
.112-40 UNC	5	4.4
.138-32 UNC	30	10.0
.164-32 UNC	35	20.0
.190-32 UNF	40	32.0
.190-24 UNC	40	35.0

4.7.3.2 Bend test (plug-in terminals only). Terminals shall be tested as specified in test-condition B (two bends). Each terminal shall be tested by applying a force of 5 pounds perpendicular to the axis of the pin within one-eighth inch of the tip of the pin.

4.7.3.3 Torque test (screw terminals only). Terminals shall be tested as specified in test-condition E, with test details as shown in table V.

Following pull, bend, and torque tests, the relays shall be examined for evidence of loosening or breaking of the terminals and other damage that could adversely affect the normal operation of the relay.

4.7.4 Insulation resistance (see 3.5.3). Relays shall be tested in accordance with method 302 of MIL-STD-202. The following details and exception shall apply:

(a) Test potential - 500 volts dc.
(b) Test equipment current - Shall not exceed 5 mA with the terminals or leads shorted.

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(c) Special conditions - Before the measurement is taken, the test voltage shall be applied for 1 minute, or until the resistance value stabilizes, whichever occurs first.
(d) Points of measurement - One test lead (the high voltage, if applicable) shall be connected to test point "a" of table VI and the other test lead shall be connected to corresponding test point "b" of table VI. Each applicable combination listed in table VI shall be tested in turn. The coil and driver input shall be energized at their rated voltages during testing of normally closed contacts.

CAUTION: During application of voltages to the relay when the relay is not operated, the terminals of the driver and coil circuits should be connected together to prevent possible damage.

TABLE VI. Test voltage points of application.

a	b
A common interconnection of the driver input terminals	A common interconnection of the driver output and coil terminals
A common interconnection of all driver and coil terminals	Case
A common interconnection of all contact terminals	Case
A common interconnection of all contact terminals	A common interconnection of all driver and coil terminals
A common interconnection of all contact terminals of each pole, in turn	A common interconnection of all other contact terminals
One terminal of each normally open contact 1/, in turn	The other terminal of the normally open contact
One terminal of each normally closed contact 1/, in turn	The other terminal of the normally closed contact

1/ Relay contacts may be tested individually or connected in parallel.

4.7.5 Dielectric withstanding voltage (see 3.5.4). Relays shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude of test voltage - In accordance with table VII.
(b) Nature of potential - 60 Hertz.
(c) Points of application of test voltage - In accordance with table VI.

TABLE VII. Dielectric withstanding voltage.

System voltage	RMS test voltage (1 minute)	1/ RMS test voltage (2-5 seconds)
Up to 50 Vdc	1,000 + 2 x voltage	120% of 1-minute value
230 Vac	1,500	1,800
230/400 Vac	1,800	2,160

1/ Optional for quality conformance inspection.

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4.7.6 Contact resistance or voltage drop (see 3.5.5). Relays shall be tested in accordance with method 307 of MIL-STD-202. The following details shall apply:

(a) Method of connection - By normal means.
(b) Test current - Rated resistive current (see 3.1).
(c) Maximum open-circuit test voltage - 6 volts dc.
(d) Number of activations prior to measurement - None.
(e) Number of test activations - The contacts shall be caused to break and then make the test current before each of 3 consecutive measurements (or, for relays rated 10 amperes or less, one reading per contact will suffice). The voltage drop shall be measured within 10 seconds after the contacts close.
(f) Number of measurements per activation - One.

For plug-in relays, if the specified voltage drop is exceeded, a substitute measurement may be made with the leads connected directly to the pins of the relay.

4.7.7 Status indication (when specified, see 3.1) (see 3.5.6). Relay contacts shall be loaded by a resistive load of 10 ±0.5 mA supplied by 50 $^{+0}_{-1}$ volt dc. The relay shall be operated and released 50 times. During the cycling the voltage drop across closed contacts and the load current with open contacts shall be measured.

4.7.8 Driver input and coil currents (see 3.5.7). The coil current and the driver input current each shall be measured with rated voltages (see 3.1) applied to the driver input and the coil circuits.

4.7.9 Operate voltage (see 3.5.8). The rated voltage (see 3.1) shall be applied to the driver input. The specified operate voltage shall be gradually applied to the relay coil circuit. A suitable device shall be used to indicate the open or closed condition of all relay contacts. The relay shall then be de-energized and a second test performed. Rated voltage shall be applied to the coil circuit. The specified driver input voltage shall be gradually applied to the driver input.

(a) For the initial test of first article inspection, relays shall be mounted in each direction of each of three mutually perpendicular planes (6 directions).
(b) For all other tests, the relays shall be mounted in the direction most likely to cause failure.

4.7.10 Release voltage (see 3.5.9). Rated operate voltage (see 3.1) shall be applied to the coil circuit and driver input. A suitable device shall be used to determine operation of the contact. Relays shall be mounted as specified in 4.7.9(a) or (b), as applicable. The coil voltage shall be gradually reduced, and the value measured at which the relay releases to the de-energized position. A second test shall then be performed. The relay coil circuit and driver input shall be re-energized with rated voltages applied. The driver input voltage shall be gradually reduced, and the value measured at which the relay releases to the de-energized position.

4.7.11 Contact bounce, operate, and release times (see 3.5.10). Photographic records of operat release, and contact bounce times shall be submitted for first article approval. Time duration of contact bounce shall be taken as the average of five consecutive readings. Current used during this test shall be not more than 100 mA. Figure 2 indicates one suggested test arrangement for obtaining the required data. Contact bounce shall be checked on both normally open and normally closed contacts. Contact bounce shall include both the initial bounce when the contacts first make and any additional bounce caused by the hammer blow of the armature going home and reflected in the contacts. Caution should be observed to be sure that the photographic record includes any of the above so-called secondary armature bounce if it exists. Simultaneity of contact action shall be observed and recorded.

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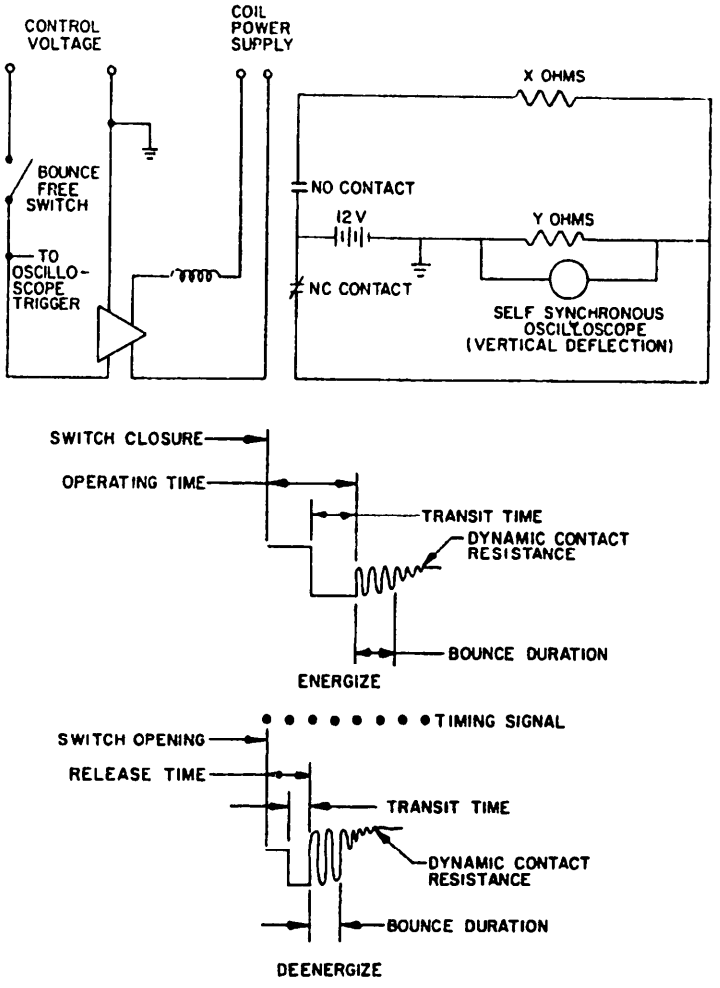


FIGURE 2. Suggested test circuit and typical record for contact bounce, operate, and release times.

4.7.12 Transient voltage (see 3.5.11). Rated voltage shall be applied to the driver output/coil circuit. The contacts shall be monitored for closed and open condition. Maximum rated release voltage shall be applied to the driver input. For a period of 1 minute at 10 ±1 ms intervals, repetitively superpose on the input a square wave transient voltage pulse of $2.00^{+.10}_{-.00}$ (unless otherwise specified) volt amplitude and 500 ±10 μs duration; overshoot shall not exceed 5 percent; rise time shall not exceed 1 μs; fall time shall not exceed 1 μs, and pulse width shall be measured at 50 percent amplitude points. Then reverse the transient voltage polarity and repeat the procedure. Repeat the preceding in its entirety, except substitute minimum driver operate voltage (see 3.1) for maximum driver release voltage.

4.7.13 Electric power dissipation (see 3.5.12). Relays shall be stabilized at 20° ±5°C. Rated operate voltages (see 3.1) shall be applied to the coil circuit and to the driver input circuit. All contacts shall be loaded with rated resistive current. The millivolt drop across each of the contacts shall be measured. The power required by the driver and the power required by the coil shall be measured within the first minute of energization. The total watts of the driver, the coil, and the contacts shall be added.

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4.7.14 Acoustical noise (see 3.5.13). Relays shall be tested, both operating and nonoperating, in accordance with method 515 of MIL-STD-810, at the levels specified (see 3.1). Nonoperating test time shall be 3 hours; operating test time shall be 30 minutes. During the test, the open contacts shall be monitored for closing and the closed contacts shall be monitored for opening. Dielectric withstanding voltage, operate voltage, and release voltage shall then be measured as specified in 4.7.5, 4.7.9, and 4.7.10, respectively.

4.7.15 Thermal shock (see 3.5.14). Relays shall be tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

(a) Special mounting - Relays shall be suspended in the test chamber by twine or other non-heat-conducting material in a plane parallel to the normal air flow. Test leads may be used for mounting; however, they shall not provide a heat sink.
(b) Test condition letter - A, except exposure time at temperature extremes shall be for 4 hours each.
(c) Measurements during cycling - None.
(d) Measurements after cycling - Operate voltage; release voltage; contact bounce, operate, and release times; insulation resistance; and dielectric withstanding voltage shall be measured as specified in 4.7.9, 4.7.10, 4.7.11, 4.7.4, and 4.7.5, respectively.

Relays shall be visually examined for breaking, cracking, chipping, and flaking of the finish and for loosening of the terminals.

4.7.16 Temperature-altitude (see 3.5.15). Relays shall be tested in accordance with method 107 of MIL-STD-202, except that the conditions of table VIII shall be used. Altitude conditions shall be simulated by testing the relays in accordance with method 105 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting - In accordance with 4.7.15(a).
(b) Other details and exceptions - See table VIII.
(c) Maximum altitude - Shall be as specified (see 3.1).
(d) Rated voltage shall be applied continuously to the driver/coil circuit. Relays shall be turned "on" and "off" by applying and removing rated voltage to the driver input, as applicable.
(e) When load is "on", contacts shall be loaded with rated resistive load current.
(f) Time duration of each step shall be 4 hours minimum.
(g) Dielectric withstanding voltage tests at altitude shall be performed at the voltage specified (see 3.1), and only between the case and a common interconnection of all terminals electrically isolated from the case.
(h) During the last 15 minutes each of steps 2, 5, 7, and 8, the following tests shall be performed:

Operate voltage	As specified in 4.7.9.
Release voltage	As specified in 4.7.10.
Contact bounce, operate, and release times	As specified in 4.7.11.
Contact resistance or voltage drop	As specified in 4.7.6.
Dielectric withstanding voltage	As specified in 4.7.5 and (g).

(i) Post-tests shall be as listed in (h), plus insulation resistance as specified in 4.7.4.

4.7.17 Continuous current (see 3.5.16). The test shall be performed at the maximum temperature and altitude specified (see 3.1). Other conditions of the test shall be in accordance with 4.4.5, except that for relays rated at 50 amperes or more, the test may be conducted with the relay mounted on a metal plate at the ambient temperature of the surrounding chamber wall. During the first 3 hours of this test, the relay shall not be energized. Normally closed contacts shall be loaded with the highest rated resistive load. At the end of the 3-hour period with no change in the ambient conditions, the operate voltage of the relays shall be determined. During the next portion of the test, the coil circuit and driver of the relay shall be energized continuously for 97 hours at their rated voltages (see 3.1). Normally open contacts shall carry the rated resistive current at any convenient

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voltage. Immediately following the operating period and with the relay still at the specified temperature, the operate voltage and release voltage shall be measured as specified in 4.7.9 and 4.7.10, respectively. Contact resistance or voltage drop; contact bounce, operate, and release times; insulation resistance; dielectric withstanding voltage; and driver input and coil currents shall then be measured at room temperature as specified in 4.7.6, 4.7.11, 4.7.4, 4.7.5, and 4.7.8, respectively, as post-tests.

TABLE VIII. Temperature-altitude test conditions.

Step	Step name	Ambient temperature (°F)	Altitude (feet)	Electrical	
				Relay condition	Load condition
1	-80°F storage (-62)	-80	Sea level	Off	---
2	-65°F warmup time (-54)	-65	Sea level	On	Off
3	Climb to max. alt., operating equipment	-65	Max. alt.	On	On
4	Frost and wet equipment warmup -10°	+14	Max. alt.	On	On
5	+203°F nonoperating 95°	+203	Sea level	Off	---
6	Normal ground operation 71°	+160	Sea level	On	On
7	50K feet compartment, normal operation	+160	50, 000	On	On
8	Max. alt. compartment, normal operation	+160	Max. alt.	On	On

4.7.18 Shock (specified pulse) (see 3.5.17). Relays shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting method - Normal mounting means.
(b) Test condition - Half-sine peak value (g), as specified (see 3.1).
(c) Electrical-load conditions - In each direction of shock, the relay shall be de-energized during two shocks and energized with rated voltage on coil and driver during one shock.
(d) Test during shock - Contacts shall be monitored for opening and closing.
(e) Examinations after shock - Operate voltage; release voltage; contact bounce, operate, and release times; and dielectric withstanding voltage shall then be measured as specified in 4.7.9, 4.7.10, 4.7.11, and 4.7.5, respectively.

4.7.19 Random vibration (see 3.5.18). Relays shall be tested in accordance with method 214 of MIL-STD-202. The following details and exceptions shall apply:

(a) During vibration testing, the open contacts shall be monitored for closing and the closed contacts shall be monitored for opening.
(b) Method of mounting - Rigidly mounted by normal mounting means.
(c) Test condition - In accordance with table IX. Duration of test shall be 22 hours in each direction. Fifty percent of the time, in each of the three mutually perpendicular planes, the relay shall be operated, and shall be released for the remaining 50 percent.
(d) Measurements after test - Operate voltage; release voltage; contact bounce, operate, and release times; and dielectric withstanding voltage shall be measured as specified in 4.7.9, 4.7.10, 4.7.11, and 4.7.5, respectively.

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TABLE IX. Random vibration equipment operating.

Frequency range (mechanical cycles per second)	Spectral density/G
20 to 80	6 dB per octave increase
80 to 200	0.5 G^2/Hz
200 to 232	9 dB per octave decrease
232 to 1,000	0.36 G^2/Hz
1,000 to 2,000	9 dB per octave decrease

4.7.20 Electromagnetic interference and compatibility (EMIC) (see 3.5.19). Relays shall be tested in accordance with method CE03 of MIL-STD-461. Testing shall be done with the relay coil and drivers continuously energized at rated voltage, also with the relay coils cycled at 50 percent duty cycle at a rate between 5 and 10 operations per minute. Contacts shall not be loaded.

4.7.21 Case-to-ground resistance (see 3.5.20). Relays shall be tested in accordance with method 303 of MIL-STD-202. The following shall apply:

(a) Points of measurement - Between the relay case and the grounding pin or terminal, or mounting panel, as applicable.

4.7.22 Compartment pressure (see 3.5.21). Relays shall be subjected, while nonoperating, to pressure test simulating the compartment leakage check pressure condition. The relays shall be placed in a pressure chamber at standard ambient conditions and the chamber internal pressure shall be raised to 27.0 pounds per square inch absolute (psia) minimum for at least 1 hour. At the end of the test, the chamber pressure shall be reduced to standard ambient conditions, and operate voltage, release voltage, and dielectric withstanding voltage (case to terminals only) shall be measured as specified in 4.7.9, 4.7.10, and 4.7.5, respectively.

4.7.23 Explosive decompression (see 3.5.22). Relays shall be subjected to explosive decompression of the surrounding air. The air pressure change shall be from 11.1 to .95 psia in a time not to exceed .01 second. Following the decompression, the operate voltage, release voltage, and dielectric withstanding voltage shall be measured as specified in 4.7.9, 4.7.10, and 4.7.5, respectively.

4.7.24 Acceleration (see 3.5.23). Relays shall be tested in accordance with method 212 of MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting of specimens - By normal mounting means.
(b) Electrical loading - The complete group of 6 acceleration cycles shall be run with the relay input and coil circuits de-energized, and shall then be repeated with the input and coil circuits energized at rated voltage.
(c) Test condition letter - A, except value as specified (see 3.1).
(d) Measurements during acceleration - Both open and closed contacts shall be monitored at all times the relays are undergoing acceleration. The operate voltage, release voltage, and dielectric withstanding voltage shall be measured as specified in 4.7.9 4.7.10, and 4.7.5, respectively.

4.7.25 Corona (see 3.5.24). Relays shall be tested as specified herein, and in accordance with ASTM D1868-61T, using 150 percent of rated rms line-to-line and line-to-neutral voltages at the maximum altitude and high temperature specified (see 3.1). Each mutually isolated circuit (see table VI) rated above 50 volts shall be independently tested.

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(a) The corona detector shall consist of an oscilloscope having, as a minimum, 100 kilohertz bandwidth with 100 millivolt-per-centimeter sensitivity with 4 to 30 millihenry rf choke connected across the oscilloscope input terminals; a vacuum system with a pressure gage accuracy of ±0.2 mmHg; and a variable ac voltage source, with a high voltage voltmeter accuracy of ±4 percent of indicated voltage.

(b) Connect a load from the common connections to the vertical input of the oscilloscope. Connect the rf choke across the oscilloscope input terminal. Connect a lead from the second (ground) oscilloscope terminal to the ground connection of the power supply. Test procedure steps shall be as follows:

(1) Adjust the sweep rate on the oscilloscope to display several cycles of the ac voltage and observe the waveform. Apply a low voltage to the test terminal, and increase the voltage to 150 percent of the rated voltage. If corona is present, the corona will appear as spikes near the peaks of the ac voltage waveform. Precautions should be taken to ensure that corona is occurring in the relays and not in the test leads, terminals, and associated apparatus. The point of breakdown, after three or four repeatable cycles, will be quite sharp and stabilized. The breakdown voltage should be observed only after stabilization. Observe any indication of corona, as exhibited by spikes or other irregularities appearing on the oscilloscope.

(2) When increasing the test voltage, the lowest voltage at which the spikes appear is the corona initiating voltage (CIV); when decreasing the test voltage, the voltage at which the spikes disappear is the corona extinction voltage (CEV). Both the CIV and CEV values shall be recorded.

(3) Units having several voltages applied to different circuits or connections will require testing of each circuit at each specified voltage.

4.7.26 Rupture (see 3.5.25) (applicable to main contacts only). Relays shall make and break specified current (see 3.1) at rated contact voltage for at least 50 cycles of operation. "On" time shall be 0.2 ±.05 second and "off" time shall be 30 ±1 seconds. Each normally-closed main contact and each normally-open main contact shall be tested. Contact resistance or voltage drop shall then be measured as specified in 4.7.6. The fuse connecting the relay case to system ground shall be tested for electrical continuity.

4.7.27 Load/endurance (see 3.5.26). Relays shall be tested for inductive, motor, resistive, and lamp loads as specified (see 3.1). Lamp load test is not required if the lamp load rating does not exceed 10 percent of the resistive load rating. The duty cycle shall be in accordance with table X. Loads, connections, and test conditions shall be in accordance with 4.4. The test shall be conducted with 50 percent of the required operating cycles performed at the maximum temperature-altitude specified. Each normally-open and each normally-closed contact shall be tested. The relay shall be subjected to the operating cycles specified (see 3.1), with loads of rated current at rated voltage using the duty cycle of table X. At the completion of the cycling, the operate voltage; release voltage; contact bounce, operate, and release times; dielectric withstanding voltage; contact resistance or voltage drop; driver input and coil currents; and insulation resistance shall be measured as specified in 4.7.9, 4.7.10, 4.7.11, 4.7.5, 4.7.6, 4.7.8, and 4.7.4, respectively. The insulation resistance test shall be the final in the series of tests.

TABLE X. Duty cycle.

Load type	Time (seconds)	
	On	Off
Inductive (ac) - - -	.5 ±.05	3.0 ±0.1
Motor (ac) - - - - -	.35 ±.09	2 ±0.1
Resistive (ac) - - -	1.5 ±1.0	1.5 ±1.0
Lamp (ac) - - - - -	2 ±.05	6 ±3

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4.7.27.1 Inductive load. Circuits shall consist of inductive and resistive load elements connected in series. The power factor shall be 0.40 ±0.05 lagging.

4.7.27.2 Motor load. The load shall be an inrush of five times rated motor load current for 0.07 ±0.02 second, after which it shall be reduced to rated current for the remainder of the "on" period.

4.7.27.3 Resistive load, ac. Resistors used for loads shall have a power factor no lower than 0.98.

4.7.27.4 Lamp load. The load shall be an inrush current of 10 times the rated lamp load for a duration of 0.015 - 0.020 second, which shall be reduced to rated current for the remainder of the "on" period.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Relays shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Relays shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Relays shall be individually packaged in accordance with MIL-P-116, method III insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. The unit container shall conform to PPP-B-566 or PPP-B-676.

5.1.2 Level C. Relays shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged relays shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using non-metallic or tape banding only.

5.2.2 Level B. The packaged relays shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged relays shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

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5.2.4.1 Level A. Relays, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. Relays, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1, except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Relays, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. Relays conforming to this specification are intended for use in aircraft and airborne vehicles with 230/400 volt 400 Hertz electric power. This does not preclude their use in other applications. (The relay performs the functions of closing, or opening, or transferring electric circuits to effect the operation of electric equipment in the same or in another electric circuit and to effect an indication of the completion of each aforementioned function, upon energizing, or de-energizing of the relay coil.)

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of the applicable specification sheet.
(c) Military part number (see 3.1).
(d) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(e) Special marking, if required (see 5.3).

6.3 Service and storage life. The relay should provide service for a period of 3 years after a 2-year storage period, without maintenance.

6.4 Case construction and degassing procedure.

6.4.1 Case construction. The case, constructed as a gas-tight enclosure, should be completely sealed by fusion of glass or ceramic to metal or bonding of metal to metal. Relays should be purged of all air and filled with a suitable inert gas of such characteristics that the leakage rate may be determined by conventional means. Relays should be hermetically sealed within 1/2 hour after filling. The fill gas should have a dew point of at least 5°C colder than the low temperature specified. The relay should be filled to a pressure of 15 ±1 psia.

6.4.2 Degassing procedure. The following degassing procedure is suggested; however, every effort should be made to utilize the most effective procedure consistent with the state-of-the-art:

(a) Apply rated coil voltage (or current).
(b) Evacuate to less than 200 microns.
(c) Heat to maximum rated ambient temperature (see 3.1) with continued evacuation.
(d) Maintain heat and vacuum for 12 hours or longer, continuing the treatment until a maximum pressure of 80 microns is reached.
(e) Turn off heaters and maintain pressure for 4 hours.
(f) Close evacuation valve and fill chamber with the desired pressurizing gas.
(g) Seal relay before removing from chamber.

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6.5 First article inspection. Information pertaining to first article inspection of products cove by this specification should be obtained from the procuring activity for the specific contracts invol

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5945-F265)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRAC

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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